

Lecture 6

Heterogeneous Treatment Effects and Causal Machine Learning

ECON6083: Machine Learning in Economics

Today's Roadmap

#	TOPIC	FOCUS
1	Why Heterogeneity?	Limitations of ATE, policy targeting
2	Causal Trees	EMSE criterion, honest estimation
3	Causal Forests	Asymptotic normality, inference
4	GRF Framework	Generalized moment equations
5	DML Integration	R-Learner, Neyman orthogonality
6	Implementation	EconML, SHAP, policy learning

Goal: Move from average effects to precision policy

Part 1

Why Care About Heterogeneity?

From ATE to "Who Benefits?"

Lecture 5: We estimated ATE — one number for everyone

But policy makers need more:

QUESTION	ATE ANSWERS?
Does the program work on average?	Yes
Who benefits most?	No
Who should we target with limited budget?	No
Could the program actually hurt some people?	No

Why CATE Matters

Example *(illustrative)*: Job training costs \$5,000/person

- ATE = \$2,000 gain → looks unprofitable
- But some gain \$10,000, others \$0
- Target the winners only → profitable!

We need: $\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x]$ — the **CATE**

CATE: Conditional Average Treatment Effect

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x]$$

x = pre-treatment characteristics (age, income, education, ...)

Examples:

- Drug effect for 50-year-old males with high blood pressure
- Training return for workers with < high school education
- Ad effect for customers who visited the website 3+ times

Key shift: From "does it work?" → "for whom does it work?"

Why Heterogeneity Can Reverse Conclusions

Kidney stone surgery (Charig et al. 1986):

STONE SIZE	SURGERY A	SURGERY B	BETTER
Small stones	93% success	87% success	A
Large stones	73% success	69% success	A
Combined	78% success	83% success	B !

A wins in every subgroup but loses overall — because doctors assign severe cases to A

Lesson: ATE can get the **sign** wrong. $\tau(x)$ is not a luxury — it's necessary for correct conclusions.

Why Estimating $\tau(x)$ is Hard

In supervised learning, we observe the target Y_i and can minimize $(Y_i - \hat{Y}_i)^2$

In causal inference, we **never** observe $\tau_i = Y_i(1) - Y_i(0)$

- Each person is either treated or not — we see one outcome, not both
- Cannot compute $(\tau_i - \hat{\tau}_i)^2$ to evaluate our estimate

Naive approach: Use ML to find subgroups with large $\hat{\tau} \rightarrow$ fails!

Coin flip analogy: Flip 100 coins. Someone gets 8 heads in a row. Skill? No — just noise. Search enough subgroups and you **will** find "large effects" by chance.

The winner's curse: These "discoveries" shrink to zero on new data.

The Overfitting Problem: Same Data for Selection + Estimation

This is the **same bias** we saw in Lecture 5 (DML), just in a different form:

STEP	TREES (NON-PARAMETRIC)	LASSO (LINEAR)
Select	Choose split points with large $\hat{\tau}$	Choose variables with large $\hat{\beta}$
Estimate	Report $\hat{\tau}$ at those splits	Report $\hat{\beta}$ for those variables
Problem	Selection used noise $\rightarrow$ estimate is biased	Selection used noise $\rightarrow$ estimate is biased

Result: 95% CIs actually cover only ~70% of the truth

Solution (rest of this lecture): Honest estimation + EMSE criterion

Part 2

Causal Trees and Honest Estimation

From Standard Trees to Causal Trees

Standard CART: Split to minimize outcome variance

$$\min_{\Pi} \sum_{\ell \in \Pi} \sum_{i \in \ell} (Y_i - \bar{Y}_{\ell})^2$$

Problem: Finds where Y differs, not where τ differs

Causal Tree: Split to maximize treatment effect heterogeneity

- But τ is unobserved, so we can't compute $(\hat{\tau} - \tau)^2$
- Need a feasible criterion $\rightarrow$ EMSE

EMSE: Why $\hat{\tau}^2$ Alone is Wrong

Naive idea: Pick the partition Π that maximizes $\hat{\tau}^2$ across leaves

Problem: $E[\hat{\tau}^2] = \tau^2 + \text{Var}(\hat{\tau})$

Small leaves have high $\text{Var}(\hat{\tau}) \rightarrow$ inflated $\hat{\tau}^2 \rightarrow$ tree chases noise

Fix: Subtract the variance to debias:

$$\begin{aligned} -\widehat{EMSE}_\tau(\Pi) = & \underbrace{\frac{1}{N} \sum_i \hat{\tau}^2(X_i)}_{\text{reward heterogeneity}} \\ & - \underbrace{\left(\frac{1}{N_{tr}} + \frac{1}{N_{est}} \right) \sum_{\ell \in \Pi} \hat{V}_\ell}_{\text{penalize imprecision}} \end{aligned}$$

where $\hat{V}_\ell = S_{treat}^2/N_{treat} + S_{control}^2/N_{control}$ is within-leaf variance of the DIM estimator

Rule: Only split if heterogeneity gain $>$ precision loss

EMSE: Numerical Example

Parent node: $N = 100$, $\hat{\tau}_{parent} = 0.1$, $\hat{V}_{parent} = 0.04$

Proposed split into two children:

- Left: $N = 40$, $\hat{\tau}_L = 0.3$, $\hat{V}_L = 0.06$
- Right: $N = 60$, $\hat{\tau}_R = -0.05$, $\hat{V}_R = 0.05$

Heterogeneity term: $0.4 \times 0.3^2 + 0.6 \times 0.05^2 = 0.0375$

Parent: $0.1^2 = 0.01$

Gain: $0.0375 - 0.01 = 0.0275$

Variance penalty: $0.06 + 0.05 - 0.04 = 0.07$

Net: $0.0275 - 0.005 \times 0.07 < 0$ (don't split if weight too high)

Honest Estimation

Problem: Using the same data to choose splits AND estimate $\hat{\tau}$:

- Tree selects splits where $\hat{\tau}$ looks large (partly due to noise)
- Then reports that noisy $\hat{\tau}$ as the estimate $\rightarrow$ biased

Solution: Split the sample

	SPLITTING SAMPLE S_{tr}	ESTIMATION SAMPLE S_{est}
Used for	Deciding where to split	Estimating $\hat{\tau}$ in each leaf
Noise	ε^{tr} drives structure	ε^{est} drives estimates

Since $\varepsilon^{tr} \perp \varepsilon^{est}$: $E[\hat{\tau}_\ell \mid \Pi] = \tau_\ell$ (unbiased!)

Analogy: Study from practice problems (S_{tr}), take a different exam (S_{est})

The Cost of Honesty

	BENEFIT	COST
Bias	Unbiased $\hat{\tau}$ conditional on Π	—
Inference	Valid 95% CIs (actual coverage $\approx 95\%$)	—
Efficiency	—	Half the data used for structure only
Depth	—	Shallower trees (less adaptive)

Wager & Athey: Bias reduction is essential for inference — honesty is the default

Part 3

Causal Forests

From Trees to Forests

Single tree: High variance — small data change alters entire structure

Causal Forest: Average B honest causal trees, each built on a random subsample with random feature subsets

Forest prediction at point x :

$$\hat{\tau}(x) = \sum_{i=1}^n \alpha_i(x) \cdot \hat{\tau}_i$$

Forest Weights

Weight $\alpha_i(x)$: how often does observation i land in the same leaf as x ?

$$\alpha_i(x) = \frac{1}{B} \sum_{b=1}^B \frac{\mathbb{I}(X_i \in L_b(x))}{|L_b(x)|}$$

- $L_b(x)$: leaf containing x in tree b
- $|L_b(x)|$: number of observations in that leaf
- If X_i never shares a leaf with x , then $\alpha_i(x) = 0$

Asymptotic Normality (Wager & Athey 2018)

$$\frac{\hat{\tau}(x) - \tau(x)}{\hat{\sigma}(x)} \xrightarrow{d} N(0, 1)$$

This requires three conditions on how each tree is built:

CONDITION	WHAT IT SAYS	WHY IT MATTERS	IN PRACTICE
Honesty	Split data $\neq$ estimation data	Eliminates overfitting bias	Built into GRF
Regularity	Leaves shrink as n grows	Estimates become local	Don't set <code>min_leaf_size</code> too large
Subsampling	Each tree uses $s_n \ll n$ obs	Makes trees independent enough for CLT	GRF uses $s_n = n/2$

Most important: Subsampling ($s_n/n \rightarrow 0$) is what enables CLT — see next slides

Proof Strategy: Hoeffding Decomposition

Goal: Show $\hat{\theta}(x)$ is approximately normal

Idea: Decompose $\hat{\theta}$ like ANOVA decomposes variance

$$\hat{\theta} - E[\hat{\theta}] = \underbrace{\sum_{i=1}^n h_1(Z_i)}_{\text{Main effects}} + \underbrace{\sum_{i < j} h_2(Z_i, Z_j) + \dots}_{\text{Interaction terms}}$$

where $h_1(Z_i) = E[\hat{\theta} \mid Z_i] - E[\hat{\theta}]$

Meaning of $h_1(Z_i)$: If I only reveal data point Z_i , how much does $\hat{\theta}$ shift on average? Each h_1 depends on **one** observation only.

Hájek Projection: Why It's Normal

Hájek projection: Keep only the main effects

$$\dot{T} = \sum_{i=1}^n h_1(Z_i)$$

Why "linear": Not "linear function" — means **independent, additive** terms. Each $h_1(Z_i)$ depends on one data point only.

Why normal: $h_1(Z_1), h_1(Z_2), \dots, h_1(Z_n)$ are i.i.d. random variables $\rightarrow$ CLT:

$$\frac{1}{\sqrt{n}} \sum_i h_1(Z_i) \xrightarrow{d} N(0, \sigma^2)$$

Remaining question: Can we ignore the interaction terms?

Why Interactions Vanish Under Subsampling

Key connection: Subsampling rate $s_n/n \rightarrow 0$ kills interactions

Intuition: Each tree uses s_n out of n observations. The probability that **two specific** observations i, j land in the **same** tree:

$$P(i, j \text{ in same tree}) \approx \left(\frac{s_n}{n}\right)^2 \rightarrow 0$$

If i and j rarely appear together, they cannot "interact."

Formally: $\text{Var}(\text{interactions}) = O(s_n^2/n^2) \ll O(1/n) = \text{Var}(\text{main effects})$

Result: $\hat{\theta} \approx \dot{T}$ (Hájek projection), so forest inherits normality:

$$\sqrt{n}(\hat{\theta}(x) - \theta(x)) \xrightarrow{d} N(0, \sigma^2)$$

Infinitesimal Jackknife for Variance

Problem: We proved $\text{Var}(\hat{\theta}) \approx \frac{1}{n} \sum_i \text{Var}(h_1(Z_i))$, but h_1 is unknown

Solution: Estimate $h_1(Z_i)$ from the forest itself (Efron 2014)

Key insight: $h_1(Z_i)$ = "marginal influence of observation i on $\hat{\theta}$ "

How to measure this: For each tree b , let $N_{bi} \in \{0, 1\}$ = whether observation i was in tree b 's subsample. Then:

$$\hat{h}_1(Z_i) = \text{Cov}_b(\hat{\theta}_b(x), N_{bi})$$

= covariance (across trees) between "tree b 's prediction" and "was i in tree b ?"

IJ variance estimator:

$$\hat{V}_{IJ}(x) = \frac{n}{B^2} \sum_{i=1}^n \hat{h}_1(Z_i)^2$$

Advantage: Computed from a single forest run — no refitting needed

Part 4

Generalized Random Forests

Beyond CATE: The GRF Framework

Causal Forest: Specific instance of general framework

Generalized Random Forests (GRF): Estimate any parameter $\theta(x)$ defined by moment equations

Examples:

- CATE: $\theta(x) = E[Y(1) - Y(0)|X = x]$
- Quantile: $\theta(x) = Q_\tau(Y|X = x)$
- IV: $\theta(x) = \text{Local IV effect}$

Unified algorithm: Same forest machinery, different moment conditions

Local Moment Equations

General form:

$$E[\psi_{\theta(x), \nu(x)}(O_i) \mid X_i = x] = 0$$

where:

- ψ is a score function
- $\theta(x)$ is target parameter
- $\nu(x)$ are nuisance parameters
- O_i are observables

Example (CATE): Score function (orthogonal form)

$$\psi = (Y_i - m(x) - \tau(x)(D_i - e(x))) \cdot (D_i - e(x))$$

where $m(x) = E[Y|X = x]$, $e(x) = E[D|X = x]$ are nuisance parameters

GRF Estimation Procedure

Step 1: Grow forest to obtain weights $\alpha_i(x)$

- Standard tree algorithm with gradient-based splitting

Step 2: Solve weighted moment equation

$$\sum_{i=1}^n \alpha_i(x) \psi(O_i, \hat{\theta}(x), \hat{\nu}(x)) = 0$$

Step 3: Invert to get $\hat{\theta}(x)$

- Often has closed-form solution
- Example: For CATE, solve weighted regression

Gradient-Based Splitting

Problem: EMSE criterion specific to CATE

- Need general criterion for arbitrary θ

Solution: Gradient-based pseudo-outcomes

Compute pseudo-outcome: For each observation i

$$\rho_i \approx -H^{-1}\psi(O_i, \hat{\theta}_{parent})$$

where H is Hessian of moment condition

Split trees: Maximize variance of ρ_i across children

- Finds directions of maximal parameter heterogeneity

GRF Example 1: CATE Score Function

Target: $\theta(x) = E[Y_i(1) - Y_i(0) \mid X_i = x]$ (conditional average treatment effect)

Define:

- $m(x) = E[Y_i \mid X_i = x]$ — conditional mean of outcome (nuisance)
- $e(x) = P(D_i = 1 \mid X_i = x)$ — propensity score (nuisance)

Score (orthogonal / R-Learner form):

$$\psi = (Y_i - m(x) - \theta(x)(D_i - e(x))) \cdot (D_i - e(x))$$

Why orthogonal: $\partial E[\psi] / \partial m = -E[D_i - e(x)] = 0$; similarly for $e(x)$. First-order errors in nuisance estimates vanish.

GRF Example 2: IV Forest — Model Setup

Problem: Treatment D_i is endogenous — correlated with unobserved U_i

Structural equation:

$$Y_i = \theta(X_i) \cdot D_i + g(X_i) + U_i$$

- $\theta(x)$: heterogeneous treatment effect (target)
- $g(x) = E[Y_i \mid X_i = x, D_i = 0]$: baseline outcome (nuisance)
- U_i : unobserved, $\text{Cov}(D_i, U_i \mid X_i) \neq 0$ (endogeneity)

Instrument Z_i satisfies:

- **Relevance:** $\text{Cov}(Z_i, D_i \mid X_i) \neq 0$ (Z predicts D)
- **Exclusion:** $E[U_i \mid X_i, Z_i] = E[U_i \mid X_i] = 0$ (Z does not directly affect Y)

GRF Example 2: IV Forest — Score Function

Define nuisance parameters:

- $\mu(x) = E[Y_i \mid X_i = x]$ — reduced-form outcome mean
- $e(x) = E[D_i \mid X_i = x]$ — treatment mean
- $p(x) = E[Z_i \mid X_i = x]$ — instrument mean (propensity of Z)

Score (residualized IV moment):

$$\psi = (Z_i - p(x)) \cdot (Y_i - \mu(x) - \theta(x)(D_i - e(x)))$$

Intuition: After partialling out X , the residualized instrument $\tilde{Z}_i = Z_i - p(x)$ is used as the "source of exogenous variation" to identify $\theta(x)$. This is the local analog of standard 2SLS.

Closed-form (weighted IV estimator at point x):

$$\hat{\theta}(x) = \frac{\sum_i \alpha_i(x) \tilde{Z}_i \tilde{Y}_i}{\sum_i \alpha_i(x) \tilde{Z}_i \tilde{D}_i}$$

GRF Example 3: Quantile Forest

Target: $\theta(x) = Q_\alpha(Y \mid X = x)$, the conditional α -quantile

Score:

$$\psi = \mathbb{I}(Y_i \leq \theta(x)) - \alpha$$

No nuisance parameters. Solves $P(Y_i \leq \theta(x) \mid X_i = x) = \alpha$.

All three examples use the same GRF forest engine — only the score ψ differs.

Part 5

Double Machine Learning Integration

The Confounding Challenge

Model:

$$Y_i = \tau(X_i) \cdot D_i + m_0(X_i) + \varepsilon_i$$

where:

- Y_i : outcome; $D_i \in \{0, 1\}$: treatment; X_i : covariates
- $\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x]$: CATE (target)
- $m_0(x) = E[Y_i(0) \mid X_i = x]$: baseline (nuisance)
- $e(x) = P(D_i = 1 \mid X_i = x)$: propensity score (nuisance)

Confounding: Why It Matters

Confounding: $e(x)$ varies with x — treatment is non-random

- Doctors assign drugs based on patient severity
- Firms select into programs based on profitability

Need: Handle high-dimensional, nonlinear $m_0(x)$ and $e(x)$

Why Not Just Control Linearly?

Naive approach: Include covariates X in forest

Problem 1: Regularization bias

- Forests shrink coefficients to reduce variance
- Bias in nuisance parameters propagates to $\hat{\tau}(x)$

Problem 2: Functional form misspecification

- Linear controls miss non-linear confounding
- High-dimensional interactions

Solution: Double Machine Learning (DML) framework

The R-Learner: Overview

Two-stage approach:

Stage 1: Residualize outcome and treatment

- Estimate $\hat{m}(x) \approx E[Y_i | X_i = x]$ using ML (e.g., Random Forest)
- Estimate $\hat{e}(x) \approx P(D_i = 1 | X_i = x)$ using ML
- Compute residuals: $\tilde{Y}_i = Y_i - \hat{m}(X_i)$, $\tilde{D}_i = D_i - \hat{e}(X_i)$

Stage 2: Run Causal Forest on residuals

- Outcome: $\tilde{Y}$
- Treatment: $\tilde{D}$

Reference: Nie & Wager (2021), "Quasi-Oracle Estimation of Heterogeneous Treatment Effects," *Biometrika*, 108(2), 299–319.

R-Learner: Stage 1 Details

Nuisance estimation with cross-fitting:

Outcome regression:

$$\hat{m}^{(-i)}(X_i) = E[Y_i|X_i] \quad (\text{trained on fold excluding } i)$$

Propensity score:

$$\hat{e}^{(-i)}(X_i) = E[D_i|X_i] \quad (\text{trained on fold excluding } i)$$

Superscript $(-i)$: Out-of-sample prediction

- Crucial for avoiding overfitting bias

R-Learner: Stage 2 Objective

Minimize R-loss:

$$\hat{\tau}(\cdot) = \arg \min_{\tau} \sum_{i=1}^n (\tilde{Y}_i - \tau(X_i) \tilde{D}_i)^2$$

Implementation: Causal Forest solves this implicitly

- Forest weights provide local smoothing
- Solves weighted least squares in each leaf

Local moment condition (GRF form):

$$E[\tilde{D}_i(\tilde{Y}_i - \tilde{D}_i \tau(X_i)) \mid X_i = x] = 0$$

Neyman Orthogonality: The Key Property

Orthogonality condition:

$$\frac{\partial}{\partial \eta} E[\psi(O_i; \theta_0, \eta)] \Big|_{\eta=\eta_0} = 0$$

Meaning: Score function locally insensitive to nuisance errors

- First derivative w.r.t. (m, e) is zero at truth
- Small errors in $\hat{m}, \hat{e}$ have negligible first-order effect on $\hat{\tau}$

Why it matters: Converts slow ML rates to fast parametric rates

Orthogonality: Error Decomposition

Without orthogonality:

$$\hat{\theta} - \theta_0 \approx \underbrace{(\hat{m} - m_0)}_{\text{Error 1}} + \underbrace{(\hat{e} - e_0)}_{\text{Error 2}}$$

- Rate: $O_p(n^{-1/4})$ (slow ML convergence)

With orthogonality:

$$\hat{\theta} - \theta_0 \approx (\hat{m} - m_0) \times (\hat{e} - e_0)$$

- Product of errors!
- Rate: $n^{-1/4} \times n^{-1/4} = n^{-1/2}$ (parametric!)

Result: $\sqrt{n}$ -consistency despite using slow ML methods

Why Residualization Creates Orthogonality

R-Learner score:

$$\psi = \tilde{D}_i(\tilde{Y}_i - \tilde{D}_i\tau)$$

Expand:

$$\psi = (D_i - e(X_i))(Y_i - m(X_i) - (D_i - e(X_i))\tau)$$

Check derivative w.r.t. m at truth:

$$\frac{\partial}{\partial m} E[\psi] = -E[D_i - e(X_i)] = 0$$

By unconfoundedness: $E[D|X] = e(X)$, so expectation is zero!

Similar for e : Derivative is zero

Conclusion: R-Learner score is Neyman orthogonal by construction

Convergence Rate Comparison

METHOD	NUISANCE RATE	TARGET RATE	INFERENCE
Plug-in ML	$n^{-1/4}$	$n^{-1/4}$	Invalid
DML (orthogonal)	$n^{-1/4}$	$n^{-1/2}$	Valid
Oracle (known nuisances)	—	$n^{-1/2}$	Valid

Key insight: DML achieves oracle rate despite estimating nuisances!

Practical implication: Can use aggressive ML without sacrificing inference

Part 6

Paper Deep Dive: Wager & Athey (2018)

Wager & Athey (2018): Causal Forests

"Estimation and Inference of Heterogeneous Treatment Effects using Random Forests"

Journal of the American Statistical Association, 113(523), 2018

Research question: Can random forests estimate $\tau(x)$ at any x with valid pointwise confidence intervals?

ML: Honest causal forests with subsampling + infinitesimal jackknife variance estimation

Building on: Athey & Imbens (2016), "Recursive Partitioning for Heterogeneous Causal Effects," *PNAS*
— introduced honest causal trees; single trees are too noisy.

Research Question

Policy problem: We know ATE, but who should we treat?

Examples from the paper:

- Which patients benefit from a medical treatment? (personalized medicine)
- Which job seekers benefit from a training program? (labor policy)
- Which voters are persuadable? (campaign strategy)

Statistical challenge: Existing methods either

- Estimate $\tau(x)$ but without valid inference (standard ML)
- Provide valid inference but only for pre-specified subgroups (linear interactions)

Goal: Estimate $\tau(x)$ at any x with pointwise confidence intervals

Method: Causal Forest

Algorithm:

1. Draw B subsamples of size s_n from data (without replacement)
2. On each subsample, grow an honest causal tree (split $\neq$ estimation data)
3. Predict $\hat{\tau}(x)$ = weighted average across all B trees

Conditions for Valid Inference

CONDITION	ROLE
Honesty	Unbiased leaf estimates $\rightarrow$ valid CIs
Regularity	Leaves shrink $\rightarrow$ estimates localize
Subsampling ($s_n/n \rightarrow 0$)	Trees decorrelate $\rightarrow$ CLT applies

Variance estimation: Infinitesimal Jackknife (no refitting needed)

Key Theoretical Result

Theorem (Wager & Athey 2018):

Under honesty, regularity, and subsampling conditions:

$$\frac{\hat{\tau}(x) - \tau(x)}{\hat{\sigma}(x)} \xrightarrow{d} N(0, 1)$$

What this enables:

- Pointwise 95% CI: $\hat{\tau}(x) \pm 1.96 \cdot \hat{\sigma}(x)$
- Test $H_0 : \tau(x) = 0$ for any individual x
- Compare $\tau(x_1)$ vs $\tau(x_2)$ across subgroups

Convergence rate: $\sqrt{n}$ — same as if we knew the true nuisance functions

Simulation Validation (Wager & Athey 2018)

DGP: $Y_i = D_i \cdot \tau(X_i) + X_{i1} + X_{i2}^2 + \varepsilon_i, \tau(x) = 2X_1 + X_2^2$

METRIC	TARGET	CAUSAL FOREST
Coverage	95%	93–96%
Bias	0	Negligible ($n > 2000$)
MSE	Minimize	Near-oracle rates

Adaptive CART vs Honest Causal Forest

	ADAPTIVE CART	HONEST CAUSAL FOREST
Coverage	~70%	~95%
CI width	Too narrow	Appropriate

Takeaways and Practical Guidance

When to use Causal Forest:

- $n > 1000$ (ideally > 5000 with many features)
- Goal is inference on $\tau(x)$, not just prediction
- Treatment is unconfounded (or use DML integration)

Common pitfalls:

1. Skipping honesty $\rightarrow$ invalid CIs (~70% coverage instead of 95%)
2. Declaring heterogeneity from point estimates alone $\rightarrow$ use CIs
3. Ignoring confounding $\rightarrow$ use `CausalForestDML` with residualization

Implementation: See in-class exercise (EconML, SHAP, policy targeting)

Further Reading

- Athey & Imbens (2016): "Recursive Partitioning for Heterogeneous Causal Effects" (PNAS)
- Wager & Athey (2018): "Estimation and Inference of Heterogeneous Treatment Effects using Random Forests" (JASA)
- Athey, Tibshirani & Wager (2019): "Generalized Random Forests" (Annals of Statistics)
- Chernozhukov et al. (2018): "Double/Debiased Machine Learning" (Econometrics Journal)

Data and Code

This lecture:

- Slides source: `_slides/Lecture06-Heterogeneous Treatment Effects-Slides.md`
- Exercises: `exercises/Lecture06-Exercise.ipynb`

Key packages:

- Python: `econml` (CausalForestDML), `grf` (R package)

Datasets:

- Wager & Athey (2018) simulations; JOB Corps data

Thank You!

Next Lecture: DAGs and Structural Causal Models

Reading: Athey, Tibshirani & Wager (2019), "Generalized Random Forests"